DSA 210 – Introduction to Data Science

Final Report

Caffeine, Sleep and Productivity Analysis

Prepared by: **İbrahim Sualp Aytuğ**

Student ID – 32604

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Abstract

This project analyzes the relationship between daily caffeine consumption, sleep patterns, and productivity metrics collected over a one-month period. Using statistical hypothesis testing and machine learning models, the study investigates whether already mentioned physiological factors significantly influence my productivity. The results indicate a strong positive correlation between caffeine intake and productivity, as well as a moderate positive correlation between total sleep duration and productivity. Machine learning models also support the findings from EDA analysis and Hypothesis Testing steps, that caffeine consumption has the most significant influence on productivity within this personal dataset.

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1. Motivation

Analyzing the correlation between productivity and the two factors, daily caffeine consumption and sleep pattern, was very important to me as a university student because productivity is a game-changer in academic life, especially for a person in his 20s. Spending my time more effectively, even though the activity I'm doing is not related to my academic life or career, is something that I wished to accomplish for years, and by analyzing the relationship between productivity and the previously mentioned two factors which are usually believed to be related to productivity in society, my main motivation was to be able to make more informed changes in my daily habits to improve my overall productivity and efficiency based on my findings from this study.

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2. Data Collection and Methodology

The study was conducted over a one-month period, during which 25 days of data were recorded. I collected the necessary data with the following methods:

- **Sleep Data:** I used an Apple Watch to track my daily sleep stages (REM, Light, Deep) and the total sleep duration.
- **Caffeine Intake:** I manually recorded caffeine consumption in milligrams using Google Sheets. The exact amount of caffeine consumed was calculated using the information provided on coffee package labels.
- **Productivity:** Productivity was also recorded manually via self-reports into the same Google Sheets.

3. Exploratory Data Analysis (EDA)

Data Preprocessing

Before running any analysis, we reorganized and cleaned the dataset by applying some basic operations such as deleting empty or missing columns and rows, removing the “mg” strings on the caffeine data, renaming some columns for clarity, and converting some data to numeric to manipulate and work on them easier.

Summary Statistics

	count	mean	std	min	25%	50%	75%	max
Caffeine_mg	25.0	114.76	113.518457	0.0	25.0	135.0	135.0	410.0
Productivity_min	25.0	196.60	100.485488	60.0	105.0	180.0	300.0	360.0
TotalSleep_min	25.0	438.60	52.605450	324.0	412.0	450.0	468.0	557.0
REM_min	25.0	95.28	23.129202	49.0	85.0	95.0	107.0	145.0
Light_min	25.0	288.68	42.891064	191.0	271.0	290.0	317.0	379.0
Deep_min	25.0	51.56	12.662543	33.0	39.0	52.0	59.0	75.0

Findings from the Summary Statistics:

- The average daily caffeine intake is approximately 115 mg, with a standard deviation of 114 mg, indicating high variability. Some days had very little caffeine intake, while on other days it exceeded 400 mg.
- Total sleep time is fairly consistent, averaging 439 minutes (approximately 7.3 hours) with a standard deviation of 53 minutes.
- Sleep quality statistics:
    - REM sleep averages 95 minutes, with moderate variability.
    - Light sleep is the largest portion of sleep, averaging 289 minutes, with some days significantly higher or lower.
    - Deep sleep averages 52 minutes, being the most stable sleep stage with lower variability.

Overall, sleep times are relatively stable, whereas caffeine intake and productivity show wider fluctuations, suggesting that there could be a stronger correlation between caffeine consumption and productivity.

Patterns and Distribution (Visualization)

After processing and summarizing the dataset, we created a number of plots to have an idea about the possible relationship between productivity and the variables we were investigating.

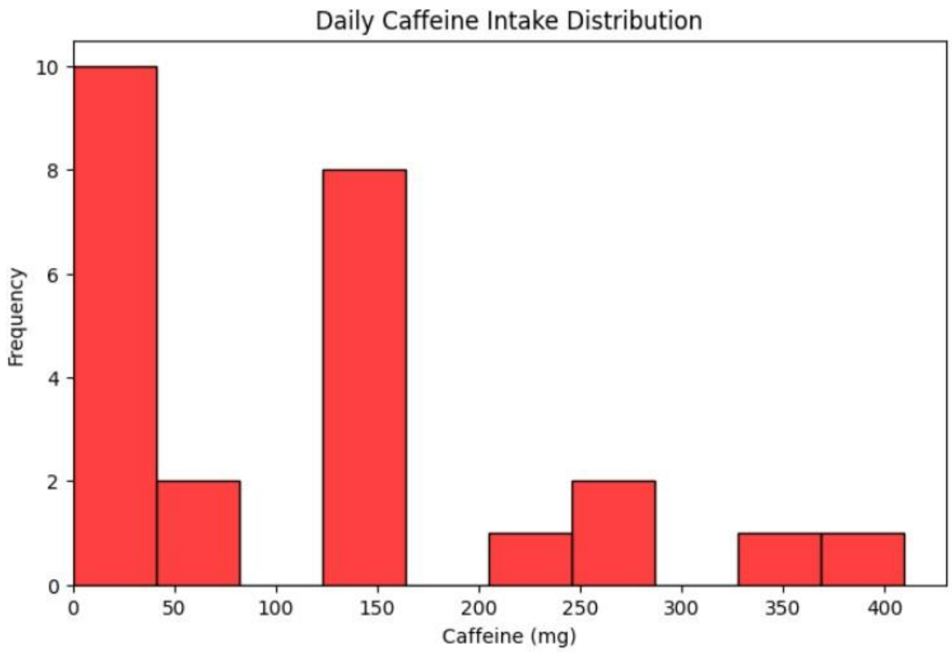

Figure 1: Daily Caffeine Intake Distribution

This histogram (Figure 1) shows the distribution of daily caffeine consumption throughout the observation period, which is about a month. The frequency of values indicates that caffeine intake was irregular. Most days fall within a lower to moderate consumption range, while a few days show remarkably high intake. The distribution is **skewed to the right**, meaning that lower caffeine consumption was more frequent, although there were occasional high values. The irregular peaks in the histogram might reflect days with higher workload, such as exam periods, which could later help us explain the observed patterns of productivity.

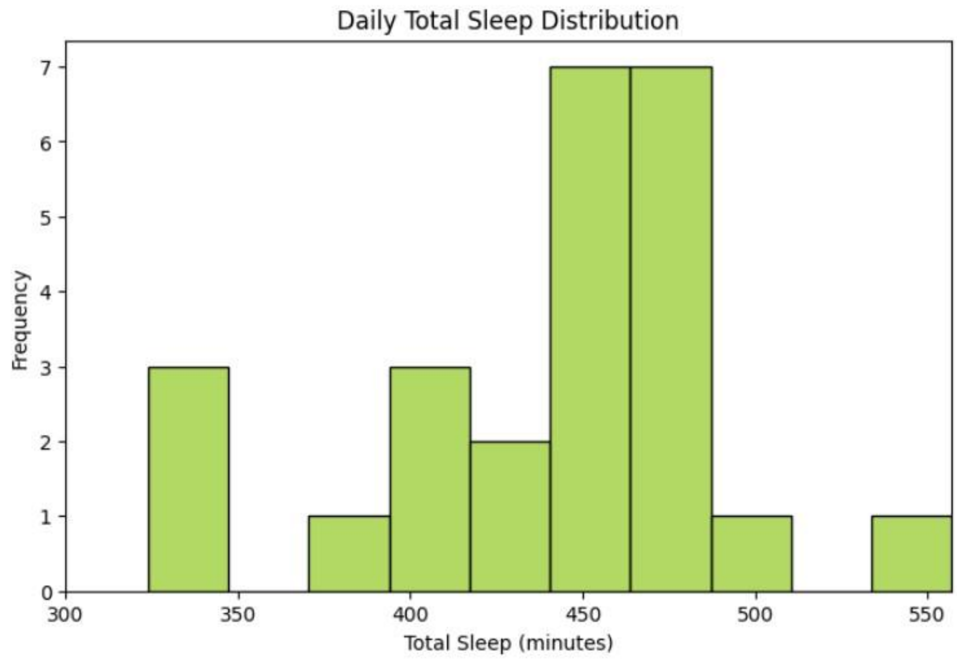

Figure 2: Daily Total Sleep Distribution

In Figure 2, we see the distribution of daily total sleep time (in minutes) recorded throughout the observation period. From the histogram, it can be observed that most days fall within the 390-480 minute range, indicating that total sleep remained relatively consistent throughout the month. There are a few days with notably shorter or longer sleep durations, suggesting limited extremes. Overall, the distribution appears roughly symmetric, supporting the stability of sleep patterns over the period.

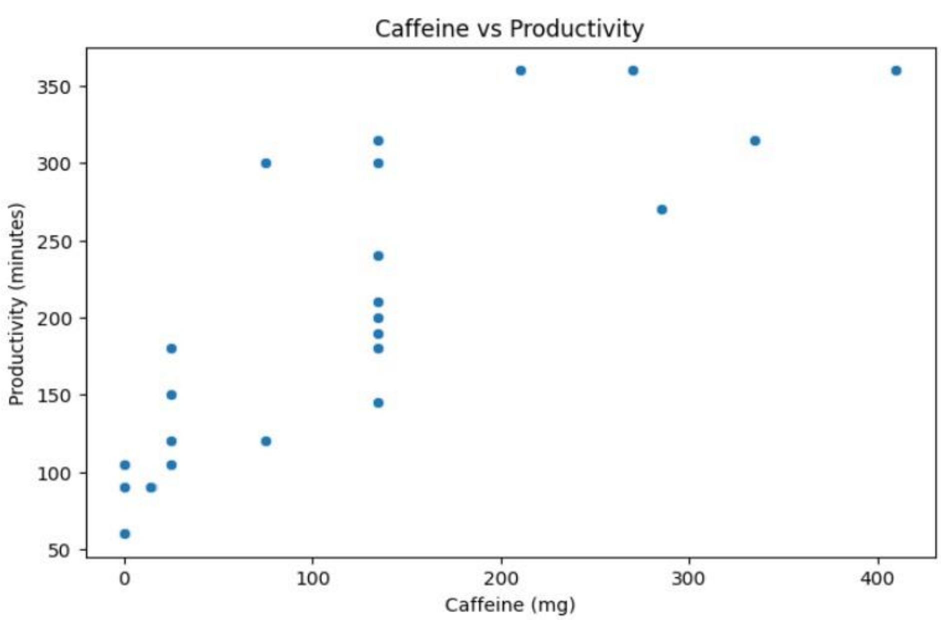

Figure 3: Caffeine vs Productivity

Based on Figure 3, such an inference can be made that higher caffeine intake generally corresponds to higher productivity, as most points trend upward while caffeine consumption increases. While most data points follow this trend, a few days show moderate or greater productivity even with lower caffeine intake, suggesting that other factors besides caffeine may also influence productivity.

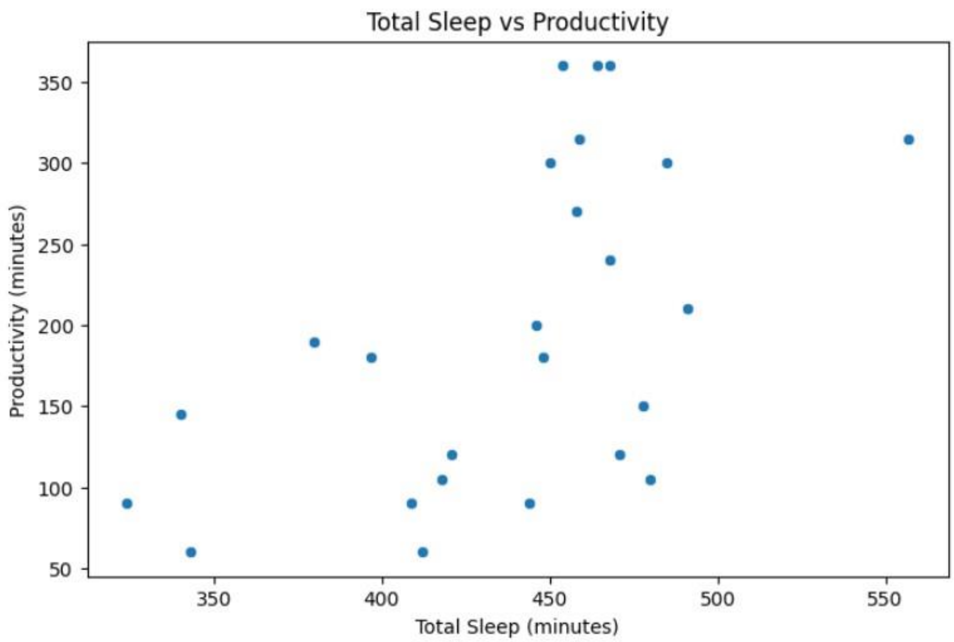

Figure 4: Total Sleep vs Productivity

The plot in Figure 4 illustrates that majority of the data points fall within 400-500 minutes range, revealing that the sleep pattern remained relatively stable in general. There appears to be a moderate positive trend: in general, days with longer sleep correspond to higher productivity. However, a notable amount of data points deviate from this pattern, suggesting that although the sleep duration may influence productivity, it is not the sole determining factor. Overall, the plot indicates that maintaining adequate sleep may support better productivity.

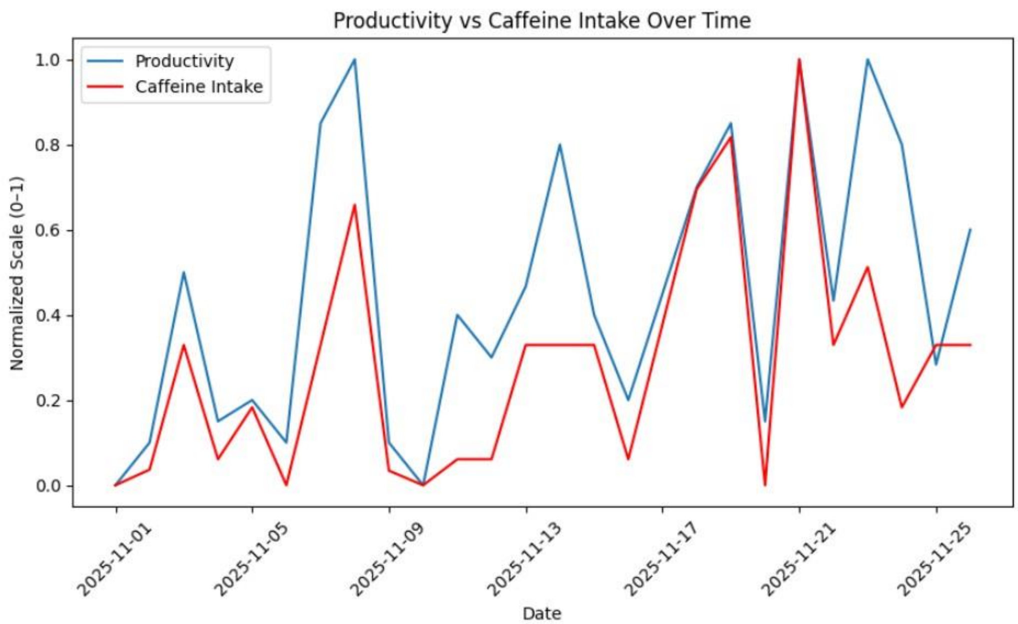

Figure 5: Productivity vs Total Sleep Time Over Time

This line graph visualizes the relationship between daily productivity and caffeine intake over time. The blue line represents productivity. while the red line illustrates caffeine intake over time. Both variables have been normalized to a 0 to 1 scale (0-1) to allow visual comparison despite their differing units.

Both lines show some fluctuations throughout the month, indicating variability in daily habits. Peaks and troughs in productivity can be observed alongside fluctuations in caffeine consumption, which helps to identify patterns or potential correlations at a glance. For example, days with higher caffeine intake often appear to coincide with higher productivity levels, suggesting a positive relationship between the two variables.

Figure 6: Productivity vs Total Sleep Time Over Time

This line graph visualizes the relationship between daily productivity and total sleep time over the observation period. The blue line represents productivity, while the green line represents total sleep time. Both of the variables are normalized to 0-1 scale again to allow visual comparison despite their differing units.

Looking at the plot, it can be observed that productivity and total sleep duration generally coincide with each other, though the alignment is less consistent than with caffeine intake, suggesting that while sleep duration may influence productivity, the effect appears moderate and not perfectly linear. The plot also shows that the sleep duration varied considerably over the month, indicating that sleep patterns were not stable, probably highly due to external factors such as upcoming exams or assignments etc.

Correlation Analysis

At this step, a correlation matrix was computed for six variables: Caffeine_mg, TotalSleep_min, REM_min, Light_min, Deep_min, and Productivity_min.

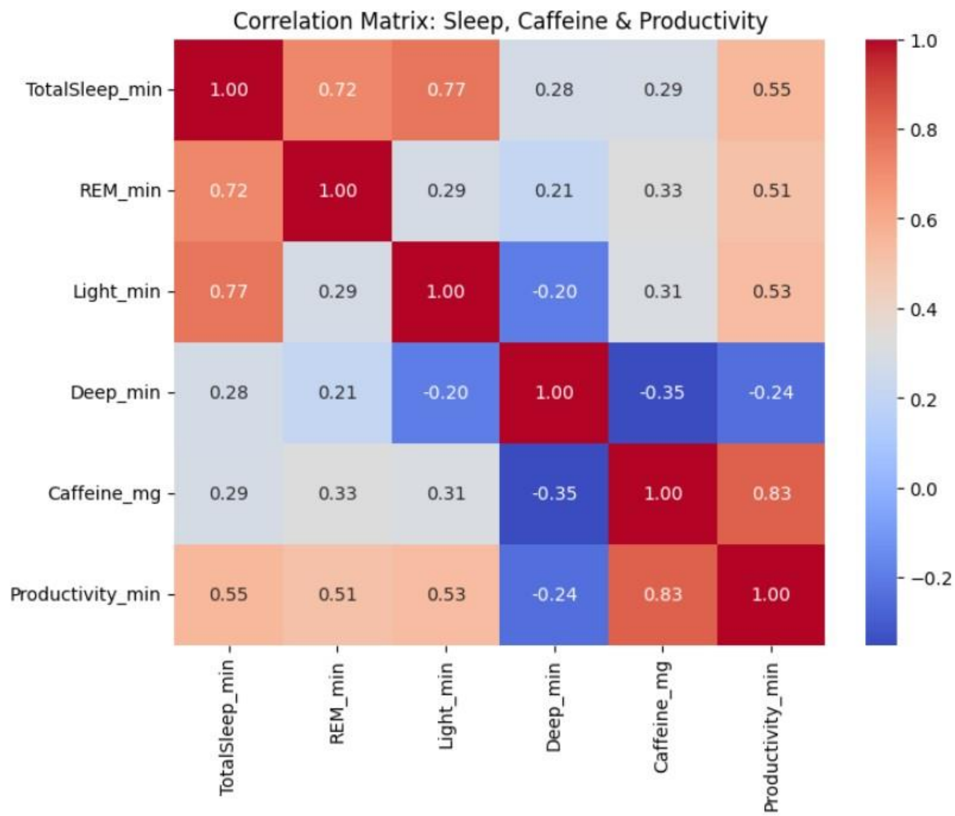

Figure 3: Correlation Matrix

Main Insights from the Heatmap:

- Productivity_min is **strongly positively correlated** with Caffeine_mg
- Productivity_min has a **moderate positive correlation** with each of TotalSleep_min, REM_min, and Light_min.
- Surprisingly, a **weak negative correlation** exists between Deep_min and Productivity_min.

This analysis gives us a solid starting point to investigate and test hypotheses about how sleep and caffeine influence productivity.

4. Hypothesis Testing

In this section, we investigated whether productivity exhibits a statistically meaningful linear association with caffeine intake and sleep pattern. First of all, we started by stating our hypotheses:

Null Hypothesis (H_0): There is no significant relationship between my productivity and my daily caffeine consumption or sleep pattern.

Alternative Hypothesis (H_a): There is a significant relationship between my productivity and my daily caffeine consumption or sleep pattern.

Pearson Correlation Test

We will calculate the Pearson correlation coefficient **r** along with the corresponding **p-value** to assess how productivity relates to caffeine consumption and, separately, to sleep patterns.

Variable Pair	r-value	p-value	Decision
Productivity / **Caffeine**	0.830	0.0000	**Reject H_0**
Productivity / Total **Sleep**	0.551	0.0043	**Reject H_0**

Test Results:

The Pearson correlation test reveals statistically significant relationships between productivity and both factors examined, in a way that supports the plots and histograms we've created.

For **caffeine consumption**, the positive correlation coefficient is $r = 0.830$ with p-value < 0.0001 (≈ 0.0000). Since the p-value is below the significance level $\alpha = 0.05$, the **null hypothesis is rejected**. This indicates a **strong positive linear association between caffeine consumption and productivity**: as caffeine intake increases, productivity tends to rise proportionally in this dataset.

For the sleep duration, the positive correlation coefficient is $r = 0.551$ with p-value $= 0.0043$. The p-value of sleep duration is also below the significance level 0.05, so we **reject the null hypothesis** again. This result suggests a **moderate positive linear association between total sleep time and productivity**, meaning higher sleep duration is generally linked with higher productivity, though the association is not as strong as that observed for caffeine.

Overall, the results support the alternative hypothesis, indicating that there is a significant relationship between my productivity and both my daily caffeine consumption and sleep patterns.

5. Machine Learning Implementation

In this phase, we will apply supervised machine learning techniques to our dataset. Our goal is to solve a Predictive Regression Task: **Predict Productivity (minutes) as a continuous variable based on physiological (caffeine intake) and behavioral inputs (sleep patterns).**

Feature Engineering Plan

In order to prepare the dataset for machine learning methods, we will:

- **Select Features**
  Extract relevant columns (Caffeine, Total Sleep, REM, Light, Deep) from the raw datasheet.
- **Clean and Preprocess**
  - Remove unnecessary strings such as "mg" and convert them to numeric.
  - Apply **Standard Scaling** to normalize features to be able to compare them with each other.

Model Plan

We will train and compare two regression models:

- **Linear Regression**
- **Random Forest Regressor**

Each model will be evaluated using:

- **MSE (Mean Squared Error)**
- **RMSE (Root Mean Squared Error)**

We will also use **K-Fold Cross-Validation** to ensure our results are robust given our limited dataset size.

We will split the dataset into **training (80%)** and **testing (20%)** sets.

- Training set size: 20 samples
- Testing set size: 5 samples

We'll also scale the features to normalize them.

Model Evaluation Results

Surprisingly, the simpler Linear Regression model outperformed the more complex Random Forest. This often occurs when the relationship affects the target variable in a direct and proportional way.

Model	RMSE	Cross-Validation Error
Linear Regression	53.56	75.65
Random Forest	72.39	64.38

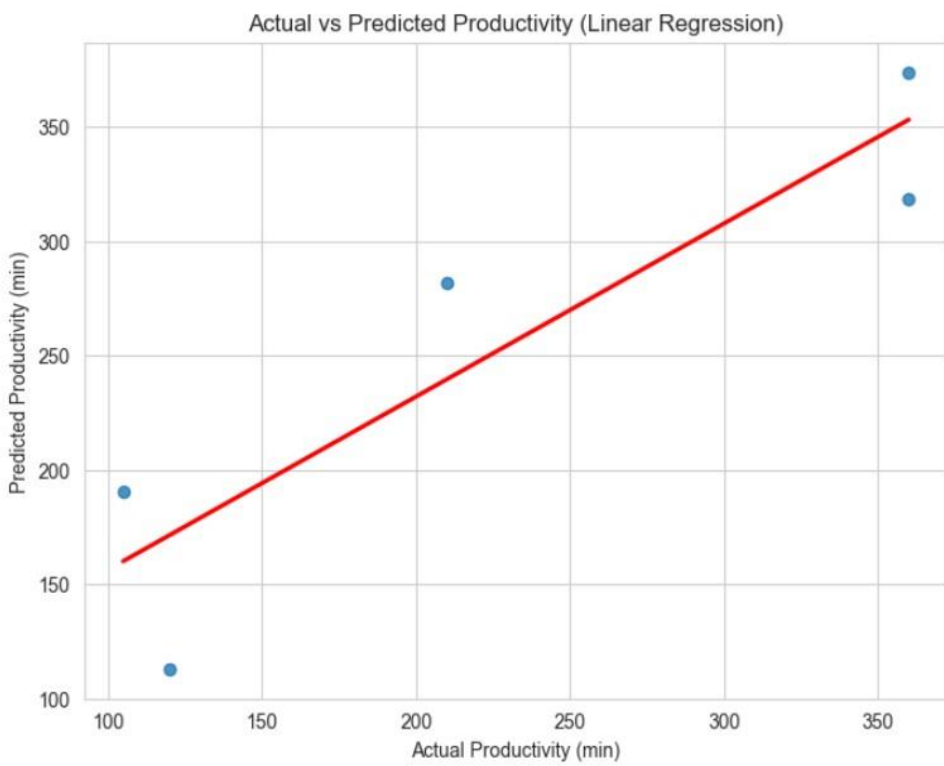

Figure 6: Linear Regression Actual vs Predicted Productivity

Feature Importance

Feature importance metrics from the models confirm that caffeine is the dominant variable, influencing the prediction more than all sleep stages combined. While REM sleep appeared as the second most relevant factor, followed by the other sleep stages, its impact was significantly smaller.

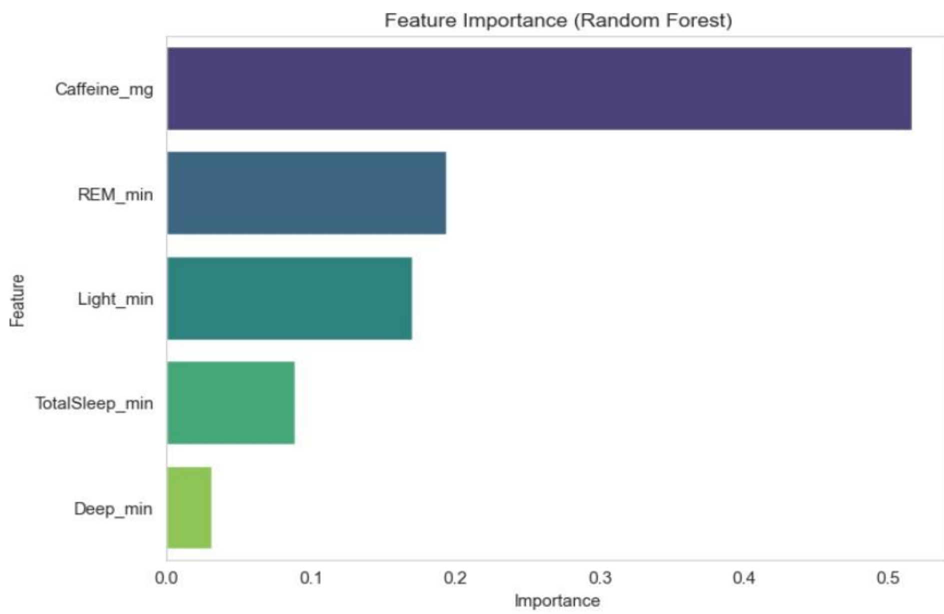

Figure 7: Feature Importance Weights (Random Forest)

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6. Limitations

- The data that collected and used to test hypotheses is my individual data. A person's productivity and the possible effects of caffeine consumption or sleep pattern may vary from person to person; therefore, the findings of this project are not generalizable.
- The dataset is limited to observations over a one-month period. The long-term effects of these factors on productivity may vary.
- The sample size is relatively small and limited, making the ML predictions less reliable for long-term generalization.

7. Future Work

Future work could focus on enriching the dataset by collecting more data continuously over a longer time period to test the hypotheses more robustly, and better train machine learning models using this expanded data to improve their prediction accuracy. Moreover, including a wider range of variables such as the daily emotional well-being, social life balances and health situations etc. could provide a more comprehensive understanding of the factors that influence productivity.

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8. Conclusion

This project applied data science techniques to identify the main factors influencing individual productivity during the observed period. The most significant result was the strong correlation between caffeine intake and productivity, showing that coffee consumption plays a major short-term role in improving daily performance. On the other hand, although sufficient sleep remains important for long-term stability, the results suggest a clear reliance on caffeine to achieve maximum productivity. Overall, these findings provide a data-driven foundation for my future lifestyle optimization and more informed decision-making.